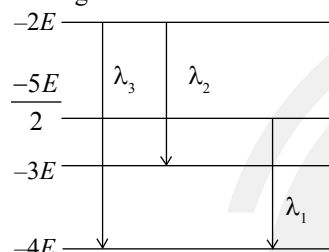


CHAPTER 12

Atoms

Alpha-Particle Scattering and Rutherford's Nuclear Model of Atom (Distance of Closest Approach and Impact Parameter, Electron Orbits)

1. Some energy levels of a molecule are shown in the figure with their wavelengths of transitions. Then: (2024 Re)



- a. $\lambda_3 > \lambda_2$, $\lambda_1 = 2\lambda_2$ b. $\lambda_3 > \lambda_2$, $\lambda_1 = 4\lambda_2$
c. $\lambda_1 > \lambda_2$, $\lambda_2 = 2\lambda_3$ d. $\lambda_2 > \lambda_1$, $\lambda_2 = 2\lambda_3$
2. Given below are two statements: (2024)

Statement I: Atoms are electrically neutral as they contain equal number of positive and negative charges.

Statement II: Atoms of each element are stable and emit their characteristic spectrum.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

- a. Statement I is correct but Statement II is incorrect.
b. Statement I is incorrect but Statement II is correct.
c. Both Statement I and Statement II are correct.
d. Both Statement I and Statement II are incorrect.
3. When an α particle of mass m moving with velocity v bombards on a heavy nucleus of charge ' Ze ', its distance of closest approach from the nucleus depends on mass: (2016 - I)
- a. $\frac{1}{m}$ b. $\frac{1}{\sqrt{m}}$ c. $\frac{1}{m^2}$ d. m

Bohr H-atom Model, Radius of Orbit, Velocity and Energy of Electrons, Wavelengths of Hydrogen Spectrum and Ionisation Potential

4. The spectral series which corresponds to the electronic transition from the levels $n_2 = 5, 6, \dots$ to the level $n_1 = 4$ is (2024 Re)
- a. Pfund series b. Brackett series
c. Lyman series d. Balmer series

5. Match List I with List II. (2024)

List I (Spectral Lines of Hydrogen for transitions from)		List II (Wavelengths (nm))	
A.	$n_2 = 3$ to $n_1 = 2$	I.	410.2
B.	$n_2 = 4$ to $n_1 = 2$	II.	434.1
C.	$n_2 = 5$ to $n_1 = 2$	III.	656.3
D.	$n_2 = 6$ to $n_1 = 2$	IV.	486.1

Choose the correct answer from the options given below:

- a. A-IV, B-III, C-I, D-II b. A-I, B-II, C-III, D-IV
c. A-II, B-I, C-IV, D-III d. A-III, B-IV, C-II, D-I
6. In hydrogen spectrum, the shortest wavelength in the Balmer series is λ . The shortest wavelength in the Bracket series is: (2023)
- a. 16λ b. 2λ c. 4λ d. 9λ
7. The radius of inner most orbit of hydrogen atom is 5.3×10^{-11} m. What is the radius of third allowed orbit of hydrogen atom? (2023)
- a. $4.77\text{\AA}$ b. $0.53\text{\AA}$ c. $1.06\text{\AA}$ d. $1.59\text{\AA}$
8. The ground state energy of hydrogen atom is -13.6 eV. The energy needed to ionize hydrogen atom from its second excited state will be: (2023-Manipur)
- a. 13.6 eV b. 6.8 eV c. 1.51 eV d. 3.4 eV
9. The angular momentum of an electron moving in an orbit of hydrogen atom is $1.5\left(\frac{h}{\pi}\right)$. The energy in the same orbit is nearly. (2023-Manipur)
- a. -1.5 eV b. -1.6 eV c. -1.3 eV d. -1.4 eV
10. Let R_1 be the radius of the second stationary and R_2 be the radius of the fourth stationary orbit of an electron in Bohr's model. The ratio $\frac{R_1}{R_2}$ is: (2022 Re)
- a. 4 b. 0.25 c. 0.5 d. 2
11. Let T_1 and T_2 be the energy of an electron in the first and second **excited states** of hydrogen atom, respectively. According to the Bohr's model of an atom, the ratio $T_1 : T_2$ is : (2022)
- a. 9 : 4 b. 1 : 4 c. 4 : 1 d. 4 : 9

12. For which one of the following, Bohr's model is **not** valid? (2020)
- Singly ionised helium atom (He^+)
 - Deuteron atom
 - Singly ionised neon atom (Ne^+)
 - Hydrogen atom
13. The total energy of an electron in the n^{th} stationary orbit of the **hydrogen atom** can be obtained by. (2020-Covid)
- $E_n = -\frac{13.6}{n^2} \text{ eV}$
 - $E_n = -\frac{1.36}{n^2} \text{ eV}$
 - $E_n = -13.6 \times n^2 \text{ eV}$
 - $E_n = \frac{13.6}{n^2} \text{ eV}$
14. The total energy of an electron in n atom in an orbit is -3.4 eV . Its kinetic and potential energies are, respectively: (2019)
- $-3.4 \text{ eV}, -3.4 \text{ eV}$
 - $-3.4 \text{ eV}, -6.8 \text{ eV}$
 - $3.4 \text{ eV}, -6.8 \text{ eV}$
 - $3.4 \text{ eV}, 3.4 \text{ eV}$
15. The ratio of kinetic energy to the total energy of an electron in a Bohr orbit of the hydrogen atom, is (2018)
- 2 : -1
 - 1 : -1
 - 1 : 1
 - 1 : -2
16. The ratio of wavelengths of the **last line** of Balmer series and the last line of Lyman series is: (2017-Delhi)
- 1
 - 4
 - 0.5
 - 2
17. If an electron in a hydrogen atom jumps from the 3rd orbit to the 2nd orbit, it **emits a photon** of wavelength λ . When it jumps from the 4th orbit to the 3rd orbit, the corresponding wavelength of the photon will be: (2016 - II)
- $\frac{20}{7}\lambda$
 - $\frac{20}{13}\lambda$
 - $\frac{16}{25}\lambda$
 - $\frac{9}{16}\lambda$
18. Given the value of Rydberg constant is 10^7 m^{-1} , the wave number of the last line of the Balmer series in hydrogen spectrum will be: (2016 - I)
- $0.025 \times 10^4 \text{ m}^{-1}$
 - $0.5 \times 10^7 \text{ m}^{-1}$
 - $0.25 \times 10^7 \text{ m}^{-1}$
 - $2.5 \times 10^7 \text{ m}^{-1}$
19. Consider 3rd orbit of He^+ (Helium) using non relativistic approach the speed of electron in this orbit will be (given $K = 9 \times 10^9$ constant $Z = 2$ and h (Planck's constant) $= 6.6 \times 10^{-34} \text{ Js}$): (2015)
- $1.46 \times 10^6 \text{ m/s}$
 - $0.73 \times 10^6 \text{ m/s}$
 - $3.0 \times 10^8 \text{ m/s}$
 - $2.92 \times 10^6 \text{ m/s}$
20. In the spectrum of hydrogen, the ratio of the longest wavelength in the Lyman series to the longest wavelength in the Balmer series is: (2015 Pre)
- 5/27
 - 4/9
 - 9/4
 - 27/5
21. Hydrogen atom in ground state is excited by a monochromatic radiation of $\lambda = 975 \text{ \AA}$. Number of spectral lines in the resulting spectrum emitted will be: (2014)
- 3
 - 2
 - 6
 - 10
22. Ratio of longest wavelengths corresponding to Lyman and Balmer series in hydrogen spectrum is: (2013)
- 9/31
 - 5/27
 - 3/23
 - 7/29

De-Broglie's Explanation of Bohr's Second Postulate of Quantisation

23. The wavelength of Lyman series of hydrogen atom appears in: (2023-Manipur)
- visible region
 - far infrared region
 - ultraviolet region
 - infrared region

Answer Key

- | | | | | | | | | | |
|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| 1. (d) | 2. (a) | 3. (a) | 4. (b) | 5. (d) | 6. (c) | 7. (a) | 8. (d) | 9. (a) | 10. (b) |
| 11. (a) | 12. (c) | 13. (b) | 14. (c) | 15. (b) | 16. (b) | 17. (a) | 18. (c) | 19. (a) | 20. (a) |
| 21. (c) | 22. (b) | 23. (c) | | | | | | | |